

Does airport dominance weaken quality competition on concentrated airline routes?

Evidence from delays and cancellations.

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May 12, 2026

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Motivation

- **Benefits and Cost of Competition** – Is competition always good and Can concentration be bad?
- **Primary Question** - How does firm behavior shift when they become the dominant option?

Past Research Findings

- **Diana, 2016** Dominance enables connectivity but introduces systemic delays (3.0–3.3 pp lower on-time rates).
- **Calzada, 2023** Dominant carriers may "internalize" congestion, scheduling conservatively to protect network integrity.
- **Chen, 2013** Quality often declines following mergers that increase dominance and reduce pressure
- **Ciliberto, 2010** Monopoly status can shift focus from satisfaction to cost minimization ?.
- **Gans, 2017** Complaint volume is **twice as sensitive** to delays when an airline controls $> 50\%$ of a market.

- **On-time is the only metric that has been used in the past to assess service**
 - This paper introduces more metrics that consumers care about: delay, arrival and cancellations.
- **Limited studies (No study**) have been conducted on COVID and the post-COVID effect.**
 - This paper presents the first multi year study on how market concentration impact service quality.

● Sample Overview

- **Source:** U.S. Domestic Flight Records (Jan 2020 – Dec 2025).
- Aggregated 40,910,273 individual flight records across 8,734 domestic routes connecting 391 airports into **Carrier-Route-Month panel**.
- **Unit of Observation:** The number of flights AA completed between ATL and LAX in January 2020, plus percentage of canceled, delays before departure, and late arrivals. ($N = 667,836$).
- **Scope:** Captures the full cycle of COVID-19 shock and recovery.

- **Dependent Variable (Y)**

- **Departure Delay Rate:** Proportion of monthly flights departing ≥ 15 mins late.
- **Arrival Delay Rate:** Proportion of monthly flights arriving later than ≥ 15 mins.
- **Cancellation Rate:** Proportion of monthly flights canceled.

- **Key Structural Predictors**

- **Market Concentration (HHI):** Sum of squared route market shares.
- **Centered Hub Share:** Carrier share at origin, centered on monthly airport mean to mitigate structural multicollinearity.
- **Schedule Dispersion (sd_dep_time):** Proxies for daily schedule volatility (scaled to 60-min units).

Theoretical background

- Model

$$Y_{irt} = \beta_1 HHI_{rt} + \beta_2 Share_{irt} + \gamma X_{irt} + \alpha_i + \delta_r + \tau_t + \epsilon_{irt} \quad (1)$$

- where

- Y_{irt} represents the delay or cancellation rate for operating carrier i on route r in month-year t .
- HHI_{rt} represents route concentration
- $Share_{irt}$ is the carrier's route market share
- X_{irt} represents a vector of operational controls.

Summary Statistics

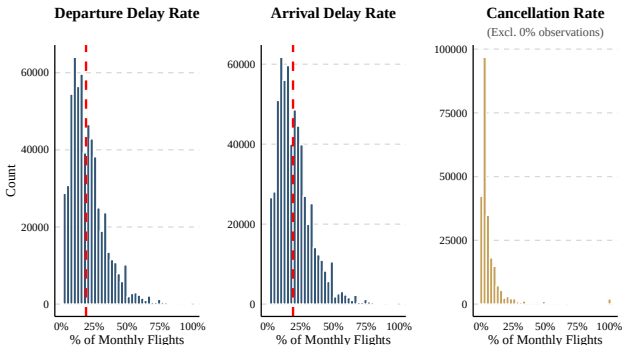
Table: Summary Statistics of the Full Sample ($N = 667,836$)

Variable	Mean	Std. Dev.	Min	Median	Max
Operational Performance (Dependent Variables)					
Departure Delay Rate	0.188	0.148	0.000	0.161	1.000
Arrival Delay Rate	0.192	0.148	0.000	0.167	1.000
Cancellation Rate	0.026	0.086	0.000	0.000	1.000
Market Structure & Controls					
HHI (Concentration)	0.704	0.274	0.138	0.657	1.000
Carrier Route Share	0.627	0.360	0.001	0.652	1.000
Carrier Hub Share	0.278	0.256	0.000	0.202	1.000
Total Route Flights	145.4	149.0	1.0	94.0	1319.0
Delay Incidents	6.37	9.04	0	3.0	248.0

Exploratory Data Analysis

- **Skewness:** Both delay rates follow a right-skewed distribution, typical of aviation operations.
- **Cancellations:** Highly concentrated near 0%, representing rare but high-impact events.

Red dashed lines indicate sample means



Airline On-Time Performance (BTS 2020 – 2025)

Exploratory Data Analysis

- **Finding:** On average, the typical route in the U.S. is highly concentrated with mean HHI of 0.7039 (7019).

Figure 4: Distribution of Market Concentration (HHI)

Visualizing the frequency of monopoly vs. competitive domestic routes

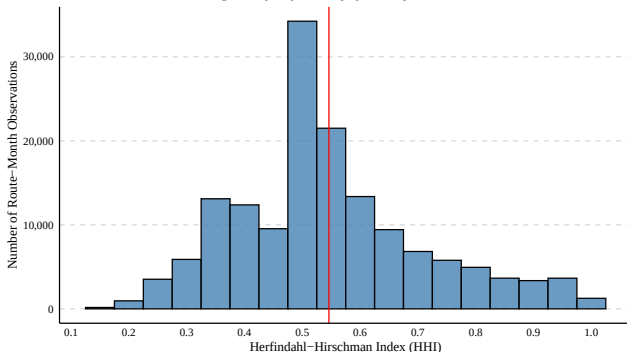


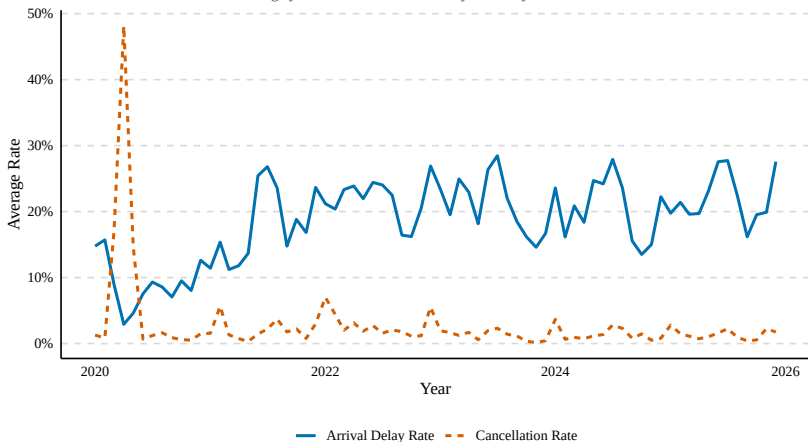
Figure: Routes with pure monopoly (HHI = 1) are not included because it distorts the plot

Exploratory Data Analysis

- **Finding:** On average, late arrivals fluctuate between 15 - 25% across all 72-months.

Figure 1: Flight Disruptions Over Time (2020–2025)

Tracking systemic shocks and recovery via delays and cancellations



Empirical Model: Fixed Effects OLS

$$\begin{aligned} \text{Delay}_{icrt} = & \beta_1 \text{HHI}_{rt} + \beta_2 \text{Share}_{icrt} + \beta_3 (\text{HHI} \times \text{Hub})_{ict} + \beta_4 (\text{HHI} \times \ln \text{Density})_{rt} + \dots \\ & \dots + \alpha_c + \gamma_r + \delta_t + \epsilon_{icrt} \end{aligned}$$

- **Fixed Effects:**

- **Route (γ_r):** Controls for time-invariant geography and airport constraints.
- **Carrier (α_c):** Accounts for airline-specific operational culture.
- **Year-Month (δ_t):** Controls for macroeconomic and seasonal shocks.

- **Identification:** Relies on *within-variation* of HHI on specific corridors across 72 monthly time-steps.

- **Robustness:**

- **Clustering:** Standard errors clustered at the **Route level**.
- **Weighting:** Weighted by *carrier total flights* to reflect the average passenger experience.

Empirical Results: Operational Performance

Table: Impact of Hub Dominance and Market Concentration

Dependent Variables: Model:	depDelay_rate (1)	arrDelay_rate (2)	cancel_rate (3)
<i>Variables</i>			
HHI	-0.0152*** (0.0033)	0.0143*** (0.0031)	-0.0107*** (0.0008)
carrierRouteShare	0.0029 (0.0030)	-0.0521*** (0.0026)	0.0159*** (0.0007)
hubShare_centered	0.0919*** (0.0075)	0.0814*** (0.0068)	-0.0027* (0.0017)
peak_dep_Count	-0.0856*** (0.0116)		-0.0849*** (0.0042)
HHI × HubShare	-0.0271** (0.0115)	-0.0198* (0.0106)	0.0142*** (0.0029)
sd_dep_time		8.64 × 10 ⁻⁵ *** (2.14 × 10 ⁻⁶)	
<i>Fixed-effects</i>			
YEAR_MONTH, Carrier, Route	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	667,440	655,816	667,440
R ²	0.48415	0.43813	0.59120
Within R ²	0.01056	0.02000	0.00391

Standard errors clustered by route; ***p<0.01, **p<0.05, *p<0.1

Part I: Key Variable Interpretations

- **The Hub Penalty:** Airport dominance (*HubShare*) is the strongest predictor of delays. A 10% increase in hub share increases baseline departure delays by **0.97 percentage points**.
- **Operational Efficiency:** Higher route share is associated with lower arrival delays ($\beta = -0.052$).
- **(Interaction):** Airlines with large shares of departing flights on a highly concentrated routes are expected to reduce their delay by 0.0271 percentage point.
 - The same cannot be said for Cancellation rate! Higher route share shows an expected increase in delays by the dominant carrier.

Conclusion

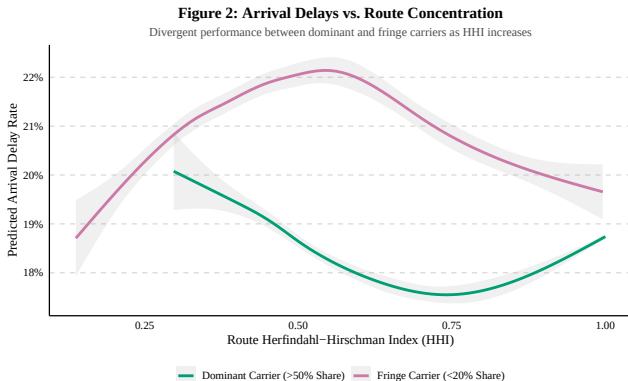


Figure: Dominant Carrier are likely to complete their flights on time versus smaller airlines.

Does airport dominance weaken quality competition? Answer: Not Exactly Yes!

- **Delays (Efficiency):** Dominance *improves* reliability in concentrated markets. Dominant hub carriers "internalize" congestion, coordinating schedules better when competition is low to avoid self-congestion.
- **Cancellations (Strategic): Yes.** The positive interaction ($\beta = 0.0142$) for cancellations suggests that in concentrated markets, dominant carriers are *more likely* to cancel flights, potentially exploiting market power when passengers have fewer alternatives.

Study Limitations

● Exogenous Shocks & Structural Breaks

- The 2020–2025 period contains extreme volatility (COVID-19 lockdowns and the 2022 recovery "meltdown").
- While fixed effects control for time, the structural relationship between HHI and delays may be fundamentally different during crisis vs. steady-state periods.

● Unobserved Environmental Variables

- **Weather:** Lack of localized METAR data. Delays attributed to concentration may be exacerbated by localized weather patterns not fully captured by route fixed effects.
- **Air Traffic Control (ATC):** FAA staffing shortages during 2023–2024 acted as a system-wide supply constraint.

● Endogeneity of Market Structure

- Airlines may exit routes *because* they are prone to delays, meaning HHI could be endogenous to operational performance.

Further research & Limitations

- There is very little month to month changes in HHI as result, it explains a small percentage of change in month variances
- Mishandled Baggage and Over-sales are metric that I would like to consider this summer.
- More comparative studies should compare the quality of service between the dominant carrier and the fringe carrier.